

A Sunspot Construction for Non-Deterministic Equilibria under Rank-Dependent Log Utility

Verification and repair note

June 25, 2026

Abstract

Wei–Xia–Zhao ask whether non-deterministic equilibrium strategies exist for continuous-time portfolio choice under rank-dependent utility. The literal printed strict-equilibrium definition, with arbitrary bounded random spike directions and strict negativity required almost surely, is too strong in any non-trivial filtration: localized spikes vanish on their complement and force a zero first-order gain there. Hence no construction can solve that literal formulation without an additional convention on perturbations. Under the support-strict convention—equivalently, under the requirement that strictness be checked on the event where the spike direction is non-zero, or for spike directions non-zero almost surely—we give a completely explicit non-deterministic equilibrium in the same Brownian incomplete-market framework. The construction uses an unspanned Brownian coordinate as a public sunspot. Before a deterministic switching time the agent has ordinary expected logarithmic utility and follows the log-Merton portfolio. After the switch the sunspot selects between two deterministic strict tail equilibria, one zero and one non-zero. The post-switch probability weighting is chosen so that the non-zero tail equilibrium is explicit, avoiding any appeal to singular-ODE existence results.

1 Status of the result

The result below is a full solution to the natural support-strict version of the non-deterministic-equilibrium question. It is not a literal solution to Definition 2.3 of Wei–Xia–Zhao [1] if that definition is read exactly as printed, because the printed definition makes strict equilibria impossible in a non-trivial filtration once spike directions may vanish on events of positive probability. This point is not special to the construction below; it also affects deterministic strategies. The support-strict convention used here is the convention under which deterministic strict equilibria in [1] are mathematically non-vacuous.

2 Model and the support-strict convention

Fix $T > 0$. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ be the usual augmentation of the natural filtration of a d -dimensional Brownian motion W . The market has one bond with constant rate r and n risky assets, $n < d$, with constant excess-return vector $\mu \in \mathbb{R}^n$ and volatility matrix $\sigma \in \mathbb{R}^{n \times d}$. Assume

$$\Sigma := \sigma \sigma^\top$$

is positive definite and $\mu \neq 0$. A portfolio-proportion process π generates wealth by

$$\frac{dX_s^\pi}{X_s^\pi} = (r + \pi_s^\top \mu) ds + \pi_s^\top \sigma dW_s.$$

We use logarithmic outcome utility $U(x) = \log x$.

For an original probability weighting function $w(t, \cdot)$ define its dual by

$$\bar{w}(t, p) = 1 - w(t, 1 - p), \quad 0 \leq p \leq 1.$$

The time- t criterion is

$$J(t, x; \pi) = \int_0^1 \log Q_{X_T^t}^t(p) \bar{w}(t, dp),$$

where Q^t denotes a conditional quantile given $\mathcal{F}_t$. This is the quantile representation used in [1], with integration against the dual weighting measure.

For $t < T$, $\varepsilon \in [0, T - t)$, and $\kappa \in L^\infty(\mathcal{F}_t; \mathbb{R}^n)$, write $\pi^{t, \varepsilon, \kappa}$ for the spike perturbation

$$\pi_s^{t, \varepsilon, \kappa} = \begin{cases} \pi_s + \kappa, & s \in [t, t + \varepsilon), \\ \pi_s, & \text{otherwise.} \end{cases}$$

Definition 1 (Support-strict equilibrium). *An admissible strategy π is a support-strict equilibrium if, for every $t < T$, every $x > 0$, and every bounded $\mathcal{F}_t$ -measurable direction κ for which the spike perturbations are admissible for all sufficiently small $\varepsilon > 0$,*

$$\lim_{\varepsilon \downarrow 0} \operatorname{ess\,sup}_{0 < \varepsilon_0 < \varepsilon} \frac{J(t, x; \pi^{t, \varepsilon_0, \kappa}) - J(t, x; \pi)}{\varepsilon_0} < 0 \quad \text{on } \{|\kappa| > 0\}.$$

Equivalently, if spike directions are restricted to satisfy $|\kappa| > 0$ a.s., the displayed inequality is required a.s.

Proposition 1 (Why the literal definition is ill posed). *Suppose $\mathcal{F}_t$ contains an event A with $0 < \mathbb{P}(A) < 1$. Let $\eta \in \mathbb{R}^n \setminus \{0\}$ and put $\kappa = \eta \mathbf{1}_A$. For every strategy π and every $\varepsilon > 0$, the perturbed strategy equals the original strategy on A^c . Hence, whenever the two objective values are finite,*

$$J(t, x; \pi^{t, \varepsilon, \kappa}) - J(t, x; \pi) = 0 \quad \text{on } A^c.$$

Thus the first-order quotient cannot be strictly negative a.s. Consequently the printed a.s. strict condition cannot hold for arbitrary bounded non-zero L^∞ directions in a non-trivial filtration. A supportwise convention, or an a.s.-non-zero convention for κ , is therefore necessary.

3 An explicit post-switch weighting and two tail equilibria

Set

$$\theta^2 := \mu^\top \Sigma^{-1} \mu > 0, \quad c_0 := \mathbb{E}[\xi(2\Phi(\xi) - 1)] = \frac{1}{\sqrt{\pi}},$$

where $\xi \sim N(0, 1)$ and Φ is the standard normal distribution function. The identity for c_0 follows by integration by parts:

$$\int_{-\infty}^{\infty} z(2\Phi(z) - 1)\phi(z) dz = 2 \int_{-\infty}^{\infty} \phi(z)^2 dz = \frac{1}{\sqrt{\pi}}.$$

Choose a tail length $H \in (0, T)$ such that

$$H < \frac{8}{\pi\theta^2},$$

and set $\tau := T - H$. Put

$$K := \frac{\theta^2}{8H}, \quad g(t) := \theta \sqrt{\frac{T-t}{2}} - \sqrt{K}(T-t), \quad \tau \leq t < T.$$

Then $g(t) > 0$ on $[\tau, T)$ and

$$\sup_{\tau \leq t < T} g(t) = g(\tau) = \frac{\theta\sqrt{H}}{2\sqrt{2}} < c_0.$$

Define $m(t) := g(t)/c_0 \in (0, 1)$ for $t \in [\tau, T)$.

For $0 \leq t < T$ define the original probability weighting field by

$$w(t, p) = \begin{cases} p, & 0 \leq t < \tau, \\ p + m(t)(p^2 - p), & \tau \leq t < T, \end{cases} \quad 0 \leq p \leq 1.$$

For $t \geq \tau$, $w_p(t, p) = 1 + m(t)(2p - 1) > 0$, so each $w(t, \cdot)$ is a valid probability weighting function. Its dual is

$$\bar{w}(t, p) = \begin{cases} p, & 0 \leq t < \tau, \\ p + m(t)(p - p^2), & \tau \leq t < T. \end{cases}$$

For $t \geq \tau$ the dual density is $1 + m(t)(1 - 2p)$. Hence the usual logarithmic h -function satisfies

$$\begin{aligned} h_x(t, 0) &= - \int_0^1 \Phi^{-1}(p) \{1 + m(t)(1 - 2p)\} dp \\ &= m(t)c_0 = g(t) > 0. \end{aligned}$$

This corrects the dual-weighting sign error in the initial draft: the weighting w above is the original distortion, while $\bar{w}$ is the measure appearing in the quantile integral.

Define two deterministic tail strategies on $[\tau, T)$ by

$$\varphi_t^0 := 0, \quad \varphi_t^1 := \frac{1}{2} \sqrt{\frac{T-t}{H}} \Sigma^{-1} \mu.$$

For φ^1 set

$$\Pi(t) := \int_t^T (\varphi_s^1)^\top \Sigma \varphi_s^1 ds.$$

Then

$$\Pi(t) = K(T-t)^2, \quad \mu = \left(1 + \frac{g(t)}{\sqrt{\Pi(t)}}\right) \Sigma \varphi_t^1, \quad \tau \leq t < T.$$

The second identity is the logarithmic rank-dependent first-order condition.

Lemma 1 (RDU of conditionally Gaussian log-wealth). *Fix $t \geq \tau$. Suppose that, conditional on $\mathcal{F}_t$, the log-terminal wealth has the form*

$$L = M + \sqrt{V} \xi,$$

where M and $V \geq 0$ are $\mathcal{F}_t$ -measurable and $\xi \sim N(0, 1)$ is independent of $\mathcal{F}_t$. Then, for the post-switch weighting,

$$\int_0^1 Q_L^t(p) \bar{w}(t, dp) = M - g(t)\sqrt{V}.$$

Proof. The conditional quantile of L is $Q_L^t(p) = M + \sqrt{V} \Phi^{-1}(p)$. Since

$$\int_0^1 \Phi^{-1}(p) \bar{w}(t, dp) = -h_x(t, 0) = -g(t),$$

the formula follows. Because log is increasing, this is exactly the rank-dependent value of terminal wealth under logarithmic utility. $\square$

Lemma 2 (The two deterministic tail strategies are support-strict). *On the tail interval $[\tau, T)$, with the post-switch weighting above, both φ^0 and φ^1 are support-strict equilibria.*

Proof. Fix $t \in [\tau, T)$ and a bounded $\mathcal{F}_t$ -measurable direction κ . Let

$$B := \kappa^\top \Sigma \kappa.$$

Since Σ is positive definite, $B > 0$ precisely on $\{|\kappa| > 0\}$.

First consider φ^0 . A spike of length ε changes the conditional mean of log-terminal wealth by

$$\varepsilon \left(\kappa^\top \mu - \frac{1}{2} B \right)$$

and changes the conditional variance from 0 to εB . By the preceding lemma,

$$J(t, x; (\varphi^0)^{t, \varepsilon, \kappa}) - J(t, x; \varphi^0) = \varepsilon \left(\kappa^\top \mu - \frac{1}{2} B \right) - g(t) \sqrt{\varepsilon B}.$$

Dividing by ε and taking the supremum over $0 < \varepsilon_0 < \varepsilon$ gives a quantity tending to $-\infty$ on $\{B > 0\}$. Thus φ^0 is support-strict.

Now consider φ^1 . Write $Y := \Pi(t) > 0$ and

$$I_\varepsilon := \int_t^{t+\varepsilon} (\varphi_s^1)^\top \Sigma \kappa \, ds.$$

Since φ^1 is continuous, $I_\varepsilon/\varepsilon \rightarrow (\varphi_t^1)^\top \Sigma \kappa$ as $\varepsilon \downarrow 0$. A spike changes the conditional mean of log-terminal wealth by

$$\varepsilon \kappa^\top \mu - I_\varepsilon - \frac{1}{2} \varepsilon B$$

and changes the conditional variance from Y to $Y + 2I_\varepsilon + \varepsilon B$. For all sufficiently small ε this variance is positive. Hence

$$\begin{aligned} & J(t, x; (\varphi^1)^{t, \varepsilon, \kappa}) - J(t, x; \varphi^1) \\ &= \varepsilon \kappa^\top \mu - I_\varepsilon - \frac{1}{2} \varepsilon B - g(t) \left(\sqrt{Y + 2I_\varepsilon + \varepsilon B} - \sqrt{Y} \right). \end{aligned}$$

After division by ε and passage to the limit, the first-order coefficient is

$$\kappa^\top \mu - C - \frac{1}{2} B - \frac{g(t)}{2\sqrt{Y}} (2C + B), \quad C := (\varphi_t^1)^\top \Sigma \kappa.$$

Using

$$\mu = \left(1 + \frac{g(t)}{\sqrt{Y}} \right) \Sigma \varphi_t^1,$$

this coefficient reduces to

$$-\frac{1}{2} \left(1 + \frac{g(t)}{\sqrt{Y}} \right) B,$$

which is strictly negative on $\{B > 0\} = \{|\kappa| > 0\}$. Since the difference quotients converge to this coefficient, their small- ε upper envelope has the same limit. Thus φ^1 is support-strict. $\square$

4 The sunspot equilibrium

Because $n < d$ and $\sigma\sigma^\top$ is positive definite, $\ker \sigma$ is non-trivial. Choose a unit vector $q \in \ker \sigma$ and set

$$C_t := q^\top W_t, \quad Z := \mathbf{1}_{\{C_\tau \geq 0\}}.$$

Then C is a Brownian motion orthogonal to the Brownian shocks spanned by traded returns, Z is $\mathcal{F}_\tau$ -measurable, and $\mathbb{P}(Z = 0) = \mathbb{P}(Z = 1) = 1/2$.

Define the global strategy

$$\pi_t = \begin{cases} \Sigma^{-1}\mu, & 0 \leq t < \tau, \\ Z\varphi_t^1 + (1 - Z)\varphi_t^0, & \tau \leq t < T. \end{cases}$$

It is bounded, progressively measurable, and admissible.

Lemma 3 (Localization over known events). *Let $A \in \mathcal{F}_t$. If terminal log-wealths L^0, L^1 have finite rank-dependent values at time t and*

$$L = \mathbf{1}_A L^1 + \mathbf{1}_{A^c} L^0,$$

then

$$\int_0^1 Q_L^t(p) \bar{w}(t, dp) = \mathbf{1}_A \int_0^1 Q_{L^1}^t(p) \bar{w}(t, dp) + \mathbf{1}_{A^c} \int_0^1 Q_{L^0}^t(p) \bar{w}(t, dp).$$

Proof. For each real y ,

$$\mathbb{P}(L \leq y \mid \mathcal{F}_t) = \mathbf{1}_A \mathbb{P}(L^1 \leq y \mid \mathcal{F}_t) + \mathbf{1}_{A^c} \mathbb{P}(L^0 \leq y \mid \mathcal{F}_t).$$

The conditional quantile therefore localizes on A and A^c :

$$Q_L^t(p) = \mathbf{1}_A Q_{L^1}^t(p) + \mathbf{1}_{A^c} Q_{L^0}^t(p)$$

for all continuity points of the relevant conditional quantile versions, and hence $\bar{w}(t, dp)$ -a.e. Integrating gives the claim. $\square$

Theorem 1 (Explicit non-deterministic support-strict equilibrium). *The strategy π is a non-deterministic support-strict equilibrium for the rank-dependent logarithmic portfolio problem with the weighting field w above.*

Proof. The strategy is non-deterministic because Z is non-degenerate and φ^1 differs from φ^0 on a set of positive Lebesgue measure.

Fix $t < T$, $x > 0$, and a bounded $\mathcal{F}_t$ -measurable direction κ .

If $t < \tau$, then for all sufficiently small ε the spike interval lies before τ . At such a time $w(t, p) = p$, so $J(t, x; \cdot)$ is ordinary conditional expected logarithmic utility. The future strategy after $t + \varepsilon$ is unchanged, and log-wealth is additive in log-returns. Therefore

$$\begin{aligned} J(t, x; \pi^{t, \varepsilon, \kappa}) - J(t, x; \pi) &= \mathbb{E}_t \int_t^{t+\varepsilon} \left(\kappa^\top \mu - (\Sigma^{-1} \mu)^\top \Sigma \kappa - \frac{1}{2} \kappa^\top \Sigma \kappa \right) ds + \mathbb{E}_t \int_t^{t+\varepsilon} \kappa^\top \sigma dW_s \\ &= -\frac{1}{2} \varepsilon \kappa^\top \Sigma \kappa. \end{aligned}$$

This is strictly negative after division by ε on $\{|\kappa| > 0\}$.

If $t \geq \tau$, then Z is already $\mathcal{F}_t$ -measurable. On $\{Z = 0\}$ the continuation strategy is exactly φ^0 , and on $\{Z = 1\}$ it is exactly φ^1 . The spike perturbation does not change Z . By the localization lemma and the deterministic tail lemma, the support-strict spike inequality holds on

$$\{|\kappa| > 0\} \cap \{Z = 0\} \quad \text{and} \quad \{|\kappa| > 0\} \cap \{Z = 1\}.$$

Combining the two events gives the desired inequality on $\{|\kappa| > 0\}$. $\square$

5 Conclusion

The construction repairs the original sunspot idea and verifies it directly. It corrects the dual-weighting convention, supplies explicit post-switch deterministic equilibria, proves the required branchwise localization of rank-dependent utility, and handles the pre-switch expected-log calculation. The outcome is a genuine non-deterministic equilibrium under the support-strict convention, or under the equivalent convention that admissible spike directions must be non-zero almost surely. If one insists on the literal printed a.s. condition for arbitrary non-zero L^∞ spike directions, the correct conclusion is different: the definition itself rules out strict equilibria in the Brownian filtration, so the non-deterministic question is ill posed without such a convention.

References

- [1] Jiaqin Wei, Jianming Xia, and Qian Zhao. *Time-Consistent Portfolio Selection for Rank-Dependent Utilities in an Incomplete Market*. arXiv:2409.19259, 2024.